Byungki Kwon

Ph.D. Candidate, POSTECH

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Research Interests

My research focuses on visual generative modeling, particularly video generation and interactive world models. Building on motion modeling and 3D geometry, I aim to make visual generation more controllable, consistent, and reliable.

WORK EXPERIENCE

Research Internship, Sony AI — Tokyo, Japan
Developed interactive video world models with persistent 3D state dynamics — Apr. 2026 – Oct. 2026

Research Internship, NAVER Cloud — Seoul, Korea
Developed a large-scale data curation pipeline for video foundation model training — Oct. 2025 – Apr. 2026

Research Internship, Microsoft Research Asia — Beijing, China
Developed a diffusion-based multimodal generative model for RGB-depth data (ICCV 2025) — Sep. 2024 – Mar. 2025

Education

POSTECH — Pohang, South Korea
Ph.D. in Artificial Intelligence (Advisor: Tae-Hyun Oh) — Mar. 2023 – Feb. 2027 (Expected)

POSTECH — Pohang, South Korea
M.S. in Electrical Engineering (Advisor: Tae-Hyun Oh) — Sep. 2020 – Feb. 2023

POSTECH — Pohang, South Korea
B.S. in Mechanical Engineering — Mar. 2013 – Aug. 2019

Publications

NeurIPS 2026
Early Failure Detection and Intervention in Video Diffusion Models
Byungki Kwon, Sohwi Lim, Nam Hyeon-Woo, Moon Ye-Bin, Tae-Hyun Oh [Code]

ICML 2026
Zero-Shot Rankability: Revealing Latent Ordinal Structure in Multimodal Large Language Models via Language
Nam Hyeon-Woo, Moon Ye-Bin, Sohwi Lim, Byungki Kwon, Tae-Hyun Oh [Code]

TMLR 2026
Revisiting Deep Video Motion Magnification for Real-time Applications
Ha Hyunwoo, Oh Hyun-Bin, Kim Jun-Seong, Byungki Kwon, Kim Sung-Bin, Ji-Yun Kim, Sung-Ho Bae, Tae-Hyun Oh

ICLR 2026
HDR-NSFF: High Dynamic Range Neural Scene Flow Fields
Shin Dong-Yeon, Kim Jun-Seong, Byungki Kwon, Tae-Hyun Oh [Code]

ICCV 2025
JointDiT: Enhancing RGB-Depth Joint Modeling with Diffusion Transformers
Byungki Kwon, Qi Dai, Lee Hyoseok, Chong Luo, Tae-Hyun Oh [Code]

ICCV Workshop 2025
Dynamic HDR Radiance Fields via Neural Scene Flow
Shin Dong-Yeon, Kim Jun-Seong, Byungki Kwon, Tae-Hyun Oh

AAAI 2025
Zero-shot Depth Completion via Test-time Alignment with Affine-invariant Depth Prior
Lee Hyoseok, Kyeong Seon Kim, Byungki Kwon, Tae-Hyun Oh [Code]

ECCV 2024
Learning-based Axial Video Motion Magnification
Byungki Kwon, Oh Hyun-Bin, Kim Jun-Seong, Hyunwoo Ha, Tae-Hyun Oh [Code]

ACCV 2024 — **The Devil is in the Details: Simple Remedies for Image-to-LiDAR Representation Learning**
Jo Won-jun, Byungki Kwon, Kim Ji-Yeon, Hawook Jeong, Kyungdon Joo, Tae-Hyun Oh

RA-L 2024 — **Uni-DVPS: Unified Model for Depth-Aware Video Panoptic Segmentation**
Kim Ji-Yeon, Oh Hyun-Bin, Byungki Kwon, Dahun Kim, Yongjin Kwon, Tae-Hyun Oh [Code]

TVCJ 2024 — **The Devil in the Details: Simple and Effective Optical Flow Synthetic Data Generation**
Byungki Kwon, Kim Sung-Bin, Tae-Hyun Oh

ICLR 2023 — **DFlow: Learning to Synthesize Better Optical Flow Datasets via a Differentiable Pipeline**
Byungki Kwon, Nam Hyeon-Woo, Kim Ji-Yun, Tae-Hyun Oh [Code]
Samsung HumanTech Paper Award

Awards & Honors

Best Paper Award, Asian Federation of Computer Vision, 2024.

Outstanding Reviewer Award, International Conference on Computer Vision (ICCV), 2023.

Bronze Prize (US$5,000), Samsung HumanTech Paper Award, 2023 (for DFlow, ICLR 2023).

Best Paper Award, Workshop on Image Processing and Image Understanding (IPIU), 2023.

Best Paper Award, Workshop on Image Processing and Image Understanding (IPIU), 2021.

Academic Service

Area Chair: CVPR 2027

Reviewer: CVPR 2024–2026; ECCV 2024, 2026; ICCV 2023, 2025 (Outstanding Reviewer, 2023); ICLR 2026; NeurIPS 2026; BMVC 2026

Patents

Method and Apparatus for Axial Motion Magnification in Video 2025
- U.S. Patent Application No. 19/355,924

Method for Self-Teaching Using Synchronized and Asynchronized Sensor Data 2023
- Republic of Korea Patent Application No. 10-2023-0146900

Image Change Data Generation Method and Apparatus 2023
- Republic of Korea Patent Application No. 10-2023-0113856

Motion Magnification System and Training Data Generation Method 2021
- Republic of Korea Patent Application No. 10-2021-0185003

Phase-Based Vibration Analysis Method and System 2020
- Republic of Korea Patent Application No. 10-2021-0044952